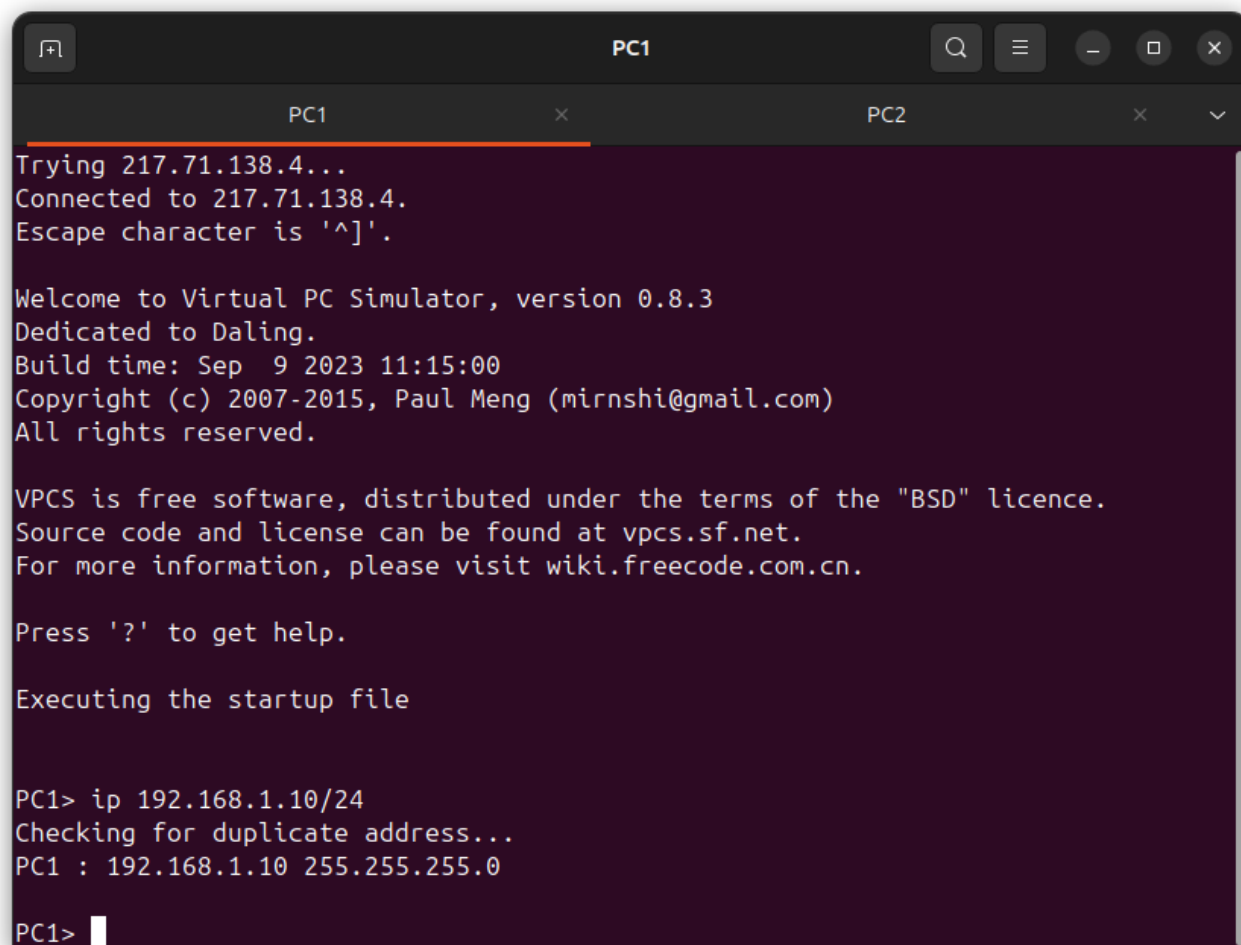


Lab_1

Сеть с коммутатором

Для назначения IP-адреса ПК 1:

```
ip 192.168.1.10/24
```



The screenshot shows a terminal window titled "PC1" with a dark background and light-colored text. The terminal output includes a connection attempt to 217.71.138.4, a welcome message for the Virtual PC Simulator version 0.8.3, and the execution of the command "ip 192.168.1.10/24". The output of this command shows the IP address 192.168.1.10 and the subnet mask 255.255.255.0. The terminal window has a standard macOS-style title bar with a search icon, a menu icon, and window control buttons (minimize, maximize, close).

```
PC1
Trying 217.71.138.4...
Connected to 217.71.138.4.
Escape character is '^]'.

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

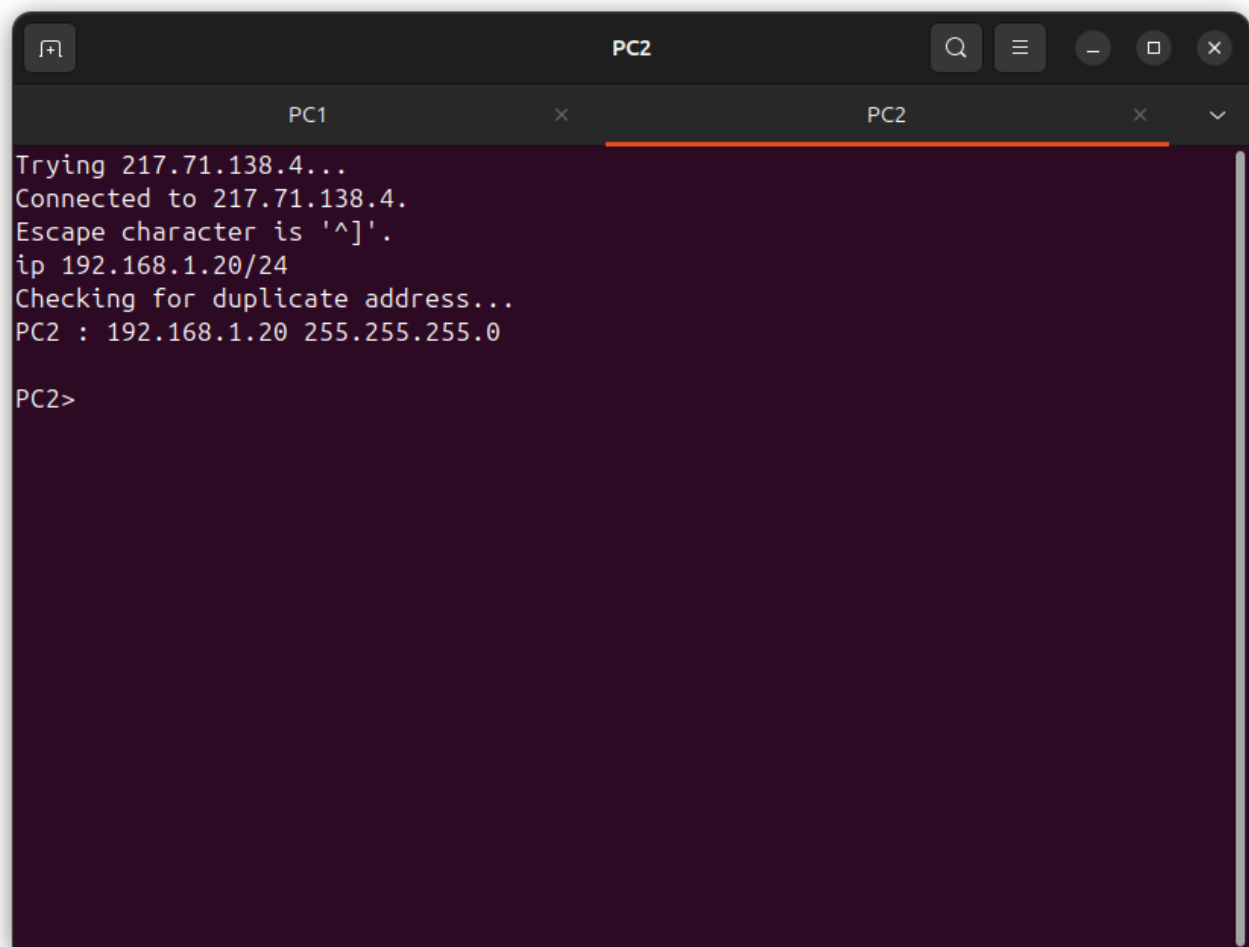
Executing the startup file

PC1> ip 192.168.1.10/24
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> 
```

Для назначения IP-адреса ПК 2:

```
ip 192.168.1.20/24
```



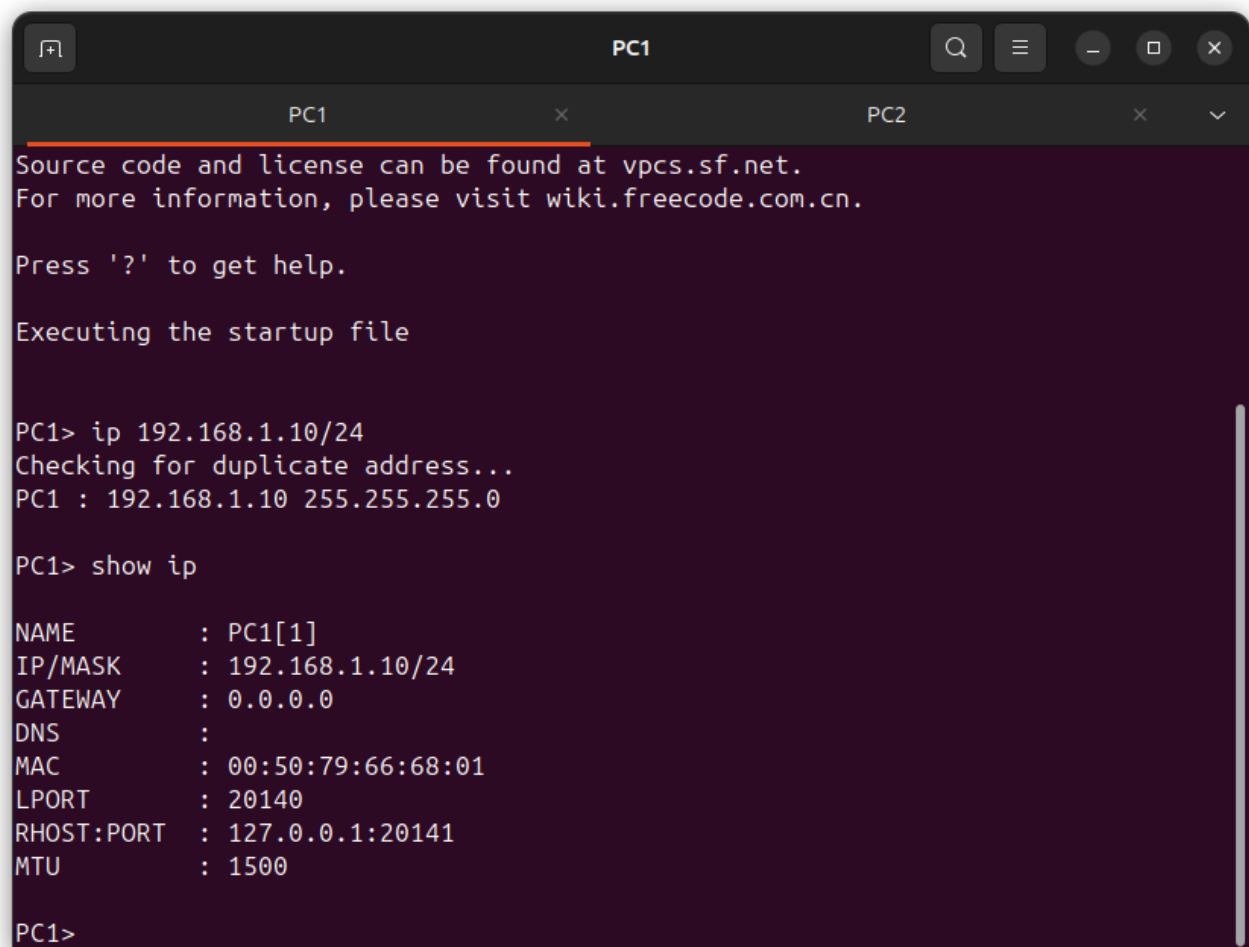
The image shows a terminal window titled 'PC2' with a dark background and light text. The terminal output shows the following sequence of commands and responses:

```
Trying 217.71.138.4...
Connected to 217.71.138.4.
Escape character is '^]'.
ip 192.168.1.20/24
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0

PC2>
```

Для проверки результата для ПК 1 и ПК 2:

```
show ip
```



PC1

Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

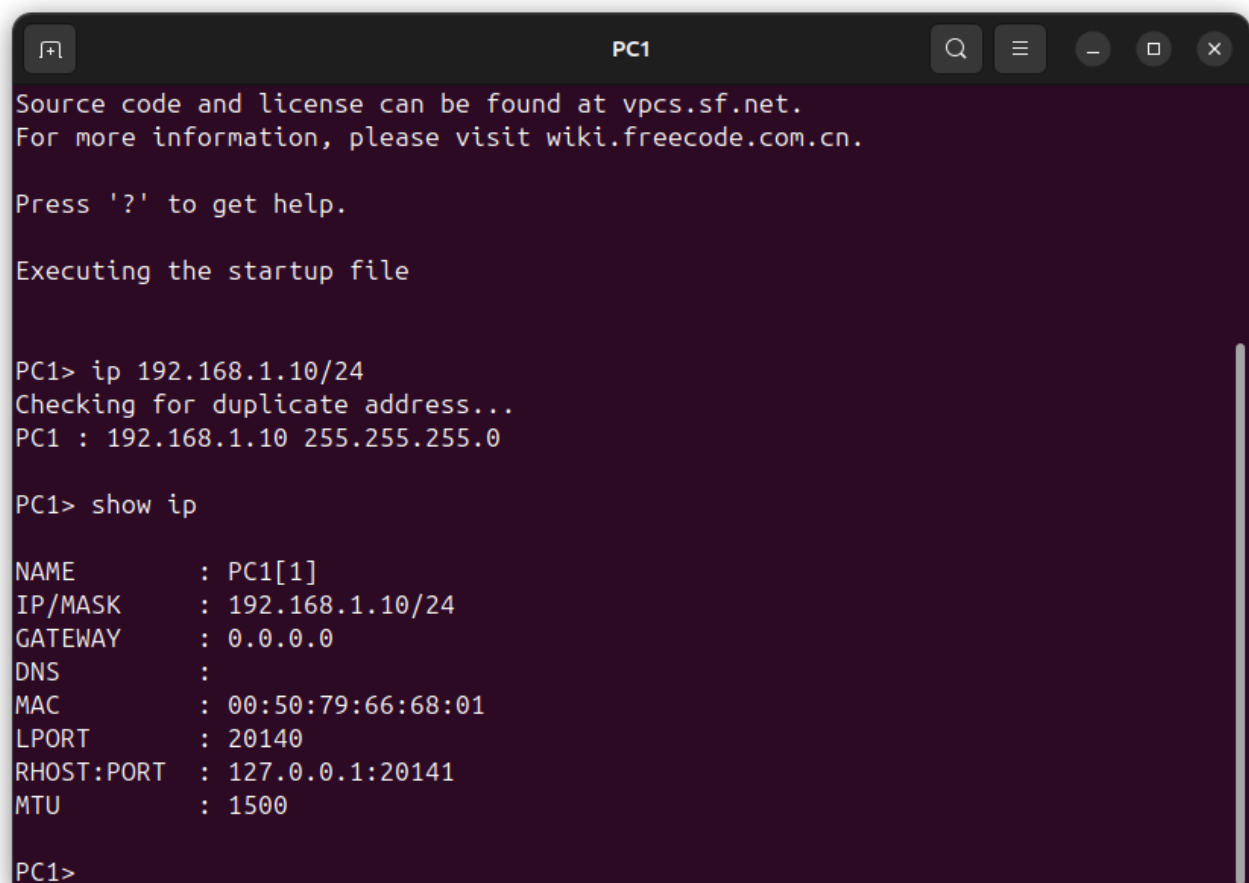
Executing the startup file

PC1> ip 192.168.1.10/24
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> show ip

NAME	: PC1[1]
IP/MASK	: 192.168.1.10/24
GATEWAY	: 0.0.0.0
DNS	:
MAC	: 00:50:79:66:68:01
LPORT	: 20140
RHOST:PORT	: 127.0.0.1:20141
MTU	: 1500

PC1>



PC1

Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

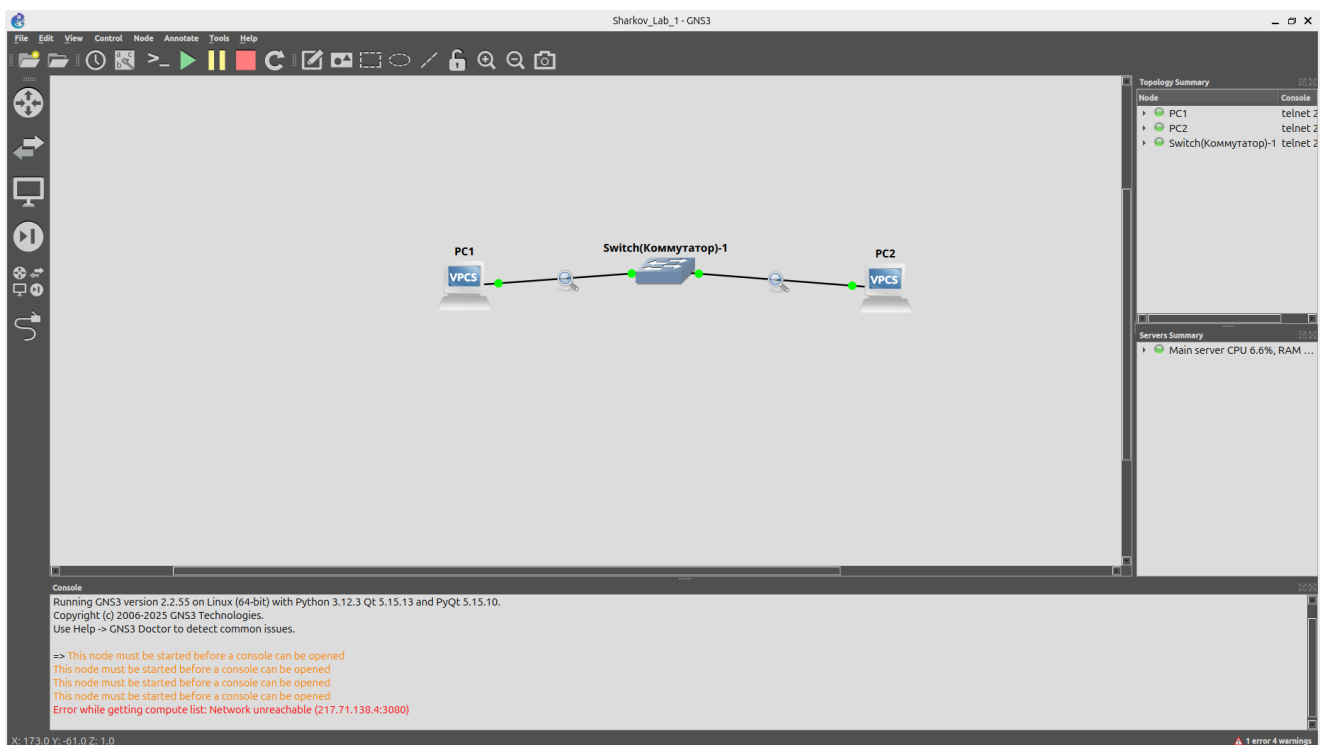
PC1> ip 192.168.1.10/24
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> show ip

NAME	: PC1[1]
IP/MASK	: 192.168.1.10/24
GATEWAY	: 0.0.0.0
DNS	:
MAC	: 00:50:79:66:68:01
LPORT	: 20140
RHOST:PORT	: 127.0.0.1:20141
MTU	: 1500

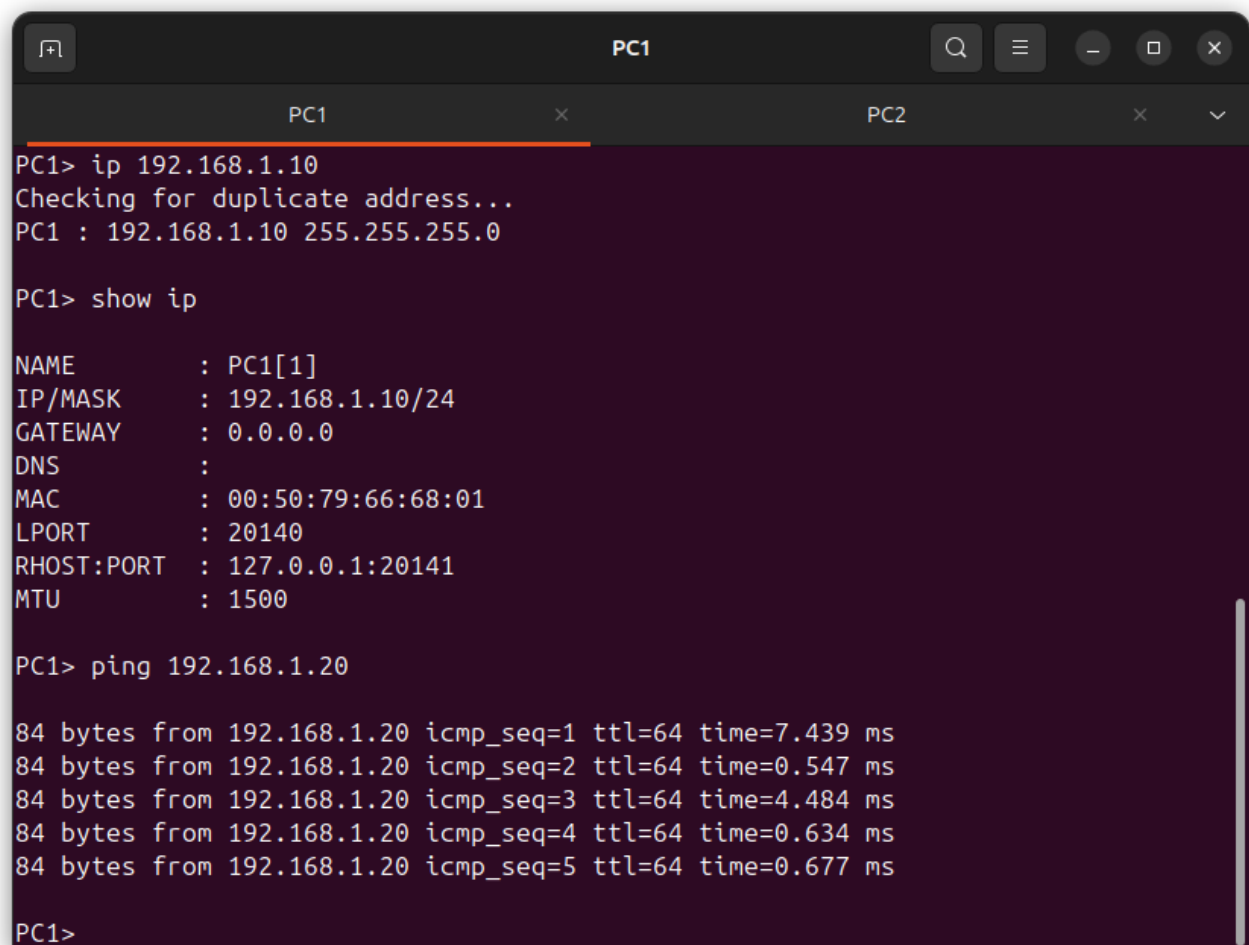
PC1>

Добавляем на соединениях интерфейс для захвата трафик через ПКМ (Start capture):



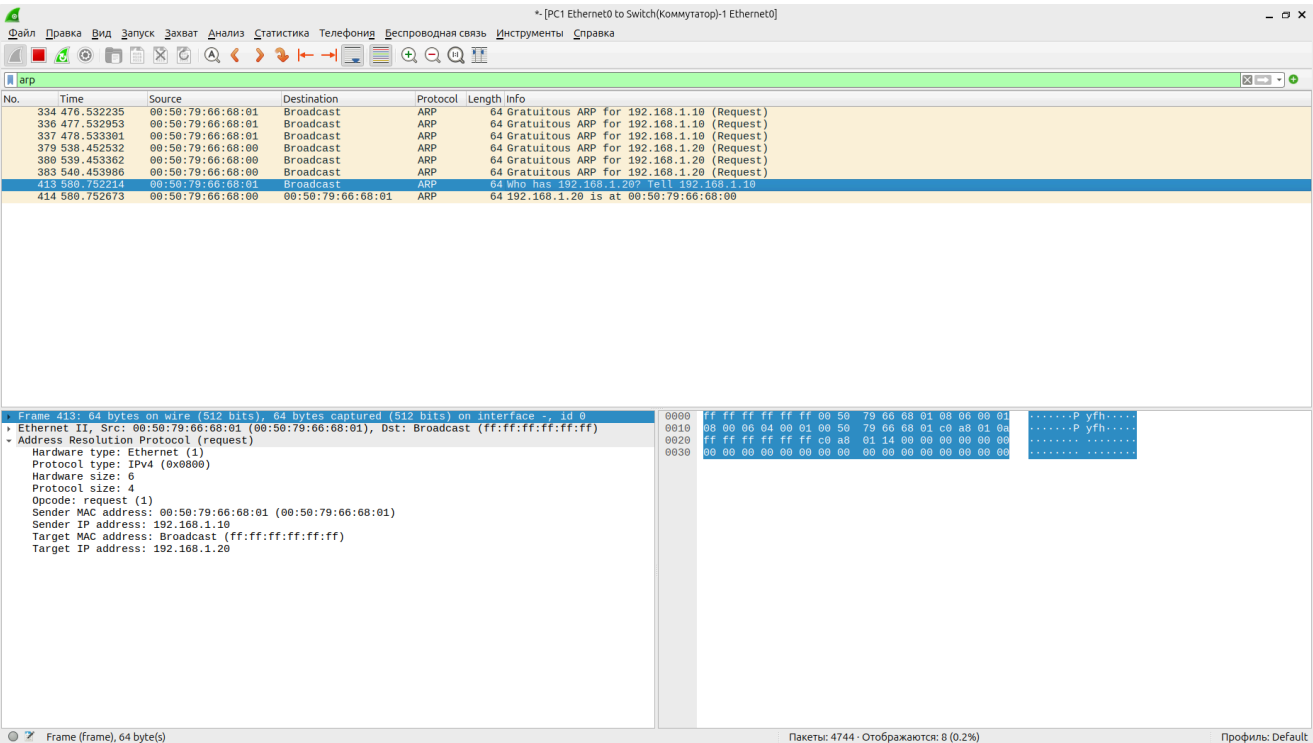
Для проверки IP-соединения с ПК 1 вызываем команду:

```
ping 192.168.1.20
```



Соединение есть

В Wireshark наблюдаем следующую картину:

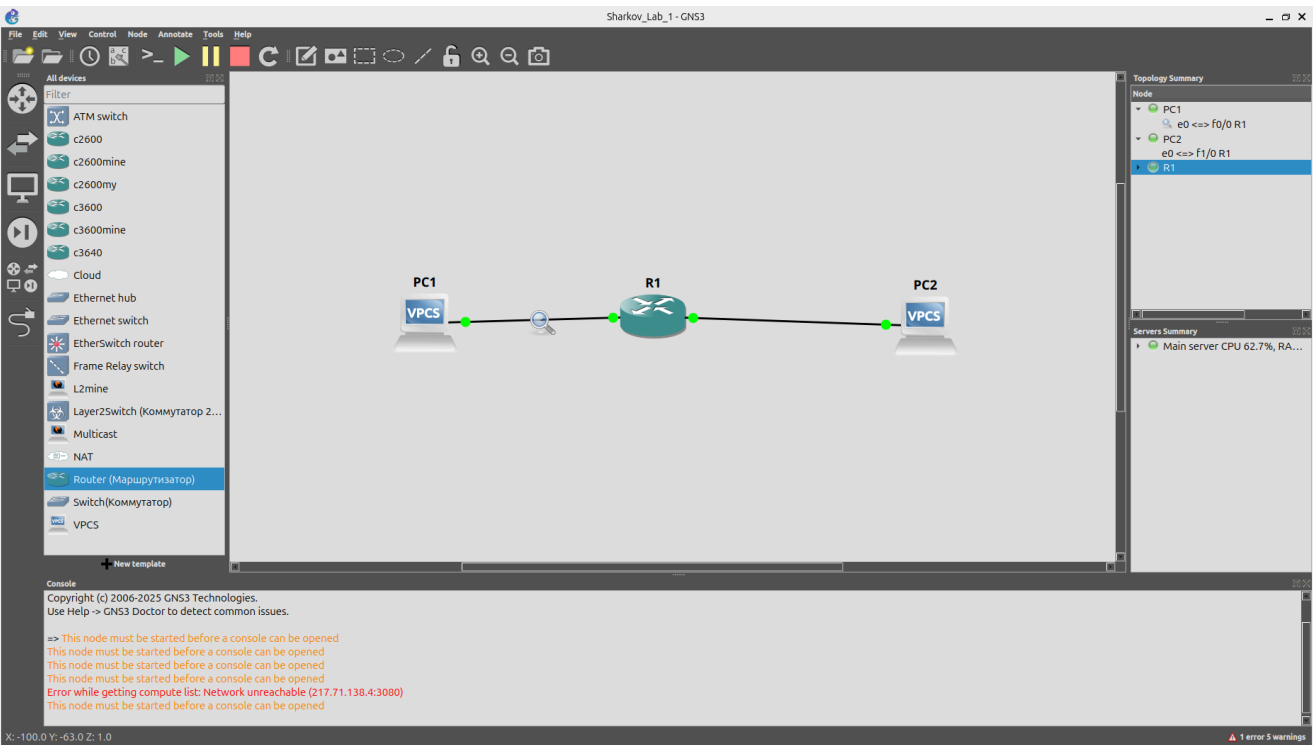


При подключении ПК 1 к сети, он отправляет 3 пакета GARP для оповещения всех в локальной сети о своём существовании в этой сети, далее, то же самое делает ПК 2. После этого видны два обычных ARP-запроса:

- Первый вызван командой ping. Происходить широковещательная рассылка запроса обладателю IP-адреса 192.168.1.20 отправить свой MAC на MAC ПК 1;
- Второй - это отсылка своего MAC ПК 2 компьютеру ПК 1.

Сеть с маршрутизатором

Меняем коммутатор на маршрутизатор:



Настраиваем маршрутизатор:

```
configure terminal      # Переход в режим глобальной конфигурации
ip routing              # Явное включение маршрутизации

interface Ethernet2/0 # Настройка нулевого интерфейса
ip address 192.168.1.1 255.255.255.0 # Назначение IP-адреса шлюзу
no shutdown            # Включить сетевую карту
exit                  # Выйти из режима редактирования интерфейса

interface Ethernet2/1 # Настройка первого интерфейса
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

exit                  # Выход из режима редактирования конфигурации
copy runnig-config startup-config # Сохранение
```

```
state to down
*Mar 1 00:00:03.551: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/2, changed
state to down
*Mar 1 00:00:03.551: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/3, changed
state to down
*Mar 1 00:00:03.551: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed st
ate to down
*Mar 1 00:00:03.555: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/1, changed st
ate to down
*Mar 1 00:00:03.555: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/2, changed st
ate to down
*Mar 1 00:00:03.559: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/3, changed st
ate to down
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface Ethernet2/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar 1 00:02:07.203: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up
*Mar 1 00:02:08.203: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed
state to up
R1(config-if)#exit
R1(config)#interface Ethernet2/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:03:18.639: %LINK-3-UPDOWN: Interface Ethernet2/1, changed state to up
*Mar 1 00:03:19.639: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed
state to up
R1(config)#exit
R1#
*Mar 1 00:03:41.071: %SYS-5-CONFIG_I: Configured from console by console
R1#copy running-config startup-config
Destination filename [startup-config]? yes
%Error copying nvram:yes (Invalid argument)
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
R1#
```

Произведена настройка интерфейсов маршрутизатора

Теперь нужно с ПК 1 пингануть шлюз 192.168.1.1 для проверки корректности настройки интерфейса роутера:

```
ping 192.162.1.1
```

```
PC1
PC1 x PC2 x R1 x
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=7.439 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=0.547 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=64 time=4.484 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=64 time=0.634 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=64 time=0.677 ms

PC1> show ip

NAME       : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20140
RHOST:PORT  : 127.0.0.1:20141
MTU         : 1500

PC1> ping 192.168.1.20

host (192.168.1.20) not reachable

PC1> ping 192.168.2.20

No gateway found

PC1> ping 192.168.2.20

No gateway found

PC1> ping 192.168.2.20

No gateway found

PC1> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=9.965 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=5.191 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=5.526 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=4.844 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=5.460 ms

PC1> 
```

Проверка прошла успешно - шлюз был обнаружен с ПК 1

То же самое для ПК 2. При пинге с ПК 2 шлюза 192.168.2.1 тоже всё успешно:

```
PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.168.1.20/24
GATEWAY     : 0.0.0.0
DNS         :
MAC        : 00:50:79:66:68:00
LPORT      : 20142
RHOST:PORT  : 127.0.0.1:20143
MTU        : 1500

PC2> ip 192.168.2.20
Checking for duplicate address...
PC2 : 192.168.2.20 255.255.255.0

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.168.2.20/24
GATEWAY     : 0.0.0.0
DNS         :
MAC        : 00:50:79:66:68:00
LPORT      : 20142
RHOST:PORT  : 127.0.0.1:20143
MTU        : 1500

PC2> ip 192.168.2.20/24
Checking for duplicate address...
PC2 : 192.168.2.20 255.255.255.0

PC2> ip 192.168.2.20/24
Checking for duplicate address...
PC2 : 192.168.2.20 255.255.255.0

PC2> ping 192.168.2.1

84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=9.360 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=5.393 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=4.963 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=5.640 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=5.144 ms

PC2> 
```

Проверка настроек роутера:

```
show ip interface brief
```

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES NVRAM    administratively down down
FastEthernet1/0          unassigned      YES NVRAM    administratively down down
Ethernet2/0               192.168.1.1     YES NVRAM    up          up
Ethernet2/1               192.168.2.1     YES NVRAM    up          up
Ethernet2/2              unassigned      YES NVRAM    administratively down down
Ethernet2/3              unassigned      YES NVRAM    administratively down down
Serial3/0                unassigned      YES NVRAM    administratively down down
Serial3/1                unassigned      YES NVRAM    administratively down down
Serial3/2                unassigned      YES NVRAM    administratively down down
Serial3/3                unassigned      YES NVRAM    administratively down down
```

В моём случае используются интерфейсы для lthernet2. Видно, что интерфейсам назначены адреса шлюзов, и они имеют проводное подключение, и включены администратором

Теперь нужно явно задать шлюз по умолчанию для ПК 1:

```
ip 192.168.1.10/24 192.168.1.1
```

```
PC1> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 20140
RHOST:PORT : 127.0.0.1:20141
MTU        : 1500
```

и ПК 2:

```
ip 192.168.2.20/24 192.168.2.1
```

```
PC2> ip 192.168.2.20/24 192.168.2.1
Checking for duplicate address...
PC2 : 192.168.2.20 255.255.255.0 gateway 192.168.2.1

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ip

NAME      : PC2[1]
IP/MASK    : 192.168.2.20/24
GATEWAY    : 192.168.2.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20142
RHOST:PORT : 127.0.0.1:20143
MTU        : 1500
```

И теперь пинг с ПК 1 ПК 2:

```
ping 192.168.2.20
```

```
PC1> ping 192.168.2.20

84 bytes from 192.168.2.20 icmp_seq=1 ttl=63 time=29.283 ms
84 bytes from 192.168.2.20 icmp_seq=2 ttl=63 time=15.511 ms
84 bytes from 192.168.2.20 icmp_seq=3 ttl=63 time=15.383 ms
84 bytes from 192.168.2.20 icmp_seq=4 ttl=63 time=15.358 ms
84 bytes from 192.168.2.20 icmp_seq=5 ttl=63 time=15.158 ms
```